Vapor Stone RAIL SYSTEMS
A Wabtec company

10 655 Henri-Bourassa Blvd., West
Saint-Laurent (Québec) Canada
H4S 1A1

Tel: 514.335.4200
Fax: 514.335.4231
www.wabtec.com

Regional Transportation District

RTD Denver EMU

Project # 7230

Door System

Software Configuration Management Plan (SCMP)

Contract Number	-	CDRL Number	-
		Customer Ref. Number	-

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	Name / Title	Signature / Date
Updated :	James Kim Software Designer	James Kim 2011-10-03
Verified :	Martin Ouellet, Eng. Software Department Supervisor	Marico Leraic for Martin Ouellet 2011-10-03
Approved :	Aldo Di Rocco Project Manager	for Aldo Di Rocco 2011-10-03

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1.2	2011-09-30	James Kim	-	Implemented comments from internal inspection I-1850
1.1	2011-09-29	James Kim	-	• Table 1-1 SCI Nomenclature is updated. • Two new members are added to the Table 2-2 SCCB Members. • Table 3-1 Baseline Establishment is updated. • Table 3-2 Configuration Items Deliverables is updated. • In Section 3.1.3.1 Documentation, the first sentence is revised to include software generic documentation..
1	2011-06-02	Jose Trejo	-	Changed date and revision for document release.
0.1	2011-05-25	Jose Trejo	-	Updated tables against Engineering deliverable status table and SPRM #2.
0.0	2011-05-10	James Kim	-	Original issue.

Preface

This SCM Plan conforms to the requirements of the Institute of Electrical and Electronic Engineers (IEEE) 828-1998 and. It also covers the requirements of the EN50128-2001 and the IEEE standard 1558-2004.

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1 INTRODUCTION

The Software Configuration Management (SCM) activities consist of all activities related to the identification, configuration control and status accounting of the SCIs as well as the reviews/audits related to SCM tasks and the configuration control of the interfaces to the system. All SCM activities identified above are further described in Figure 3-1 so that those performing and those interacting with SCM can obtain a clear understanding of the Plan.

1.1 Purpose

The purpose of this Software Configuration Management Plan (SCMP) is to define SCM responsibilities (requirements), resources, and processes used to manage and control the evolution of the Door System for the RTD Denver EMU Door System for VSRS project 7230. Preface contains compliant information as required.

This plan is intended for VSRS' software configuration manager and the development team, Rotem and Regional Transportation District Denver.

1.2 Scope

The Door System software consists of numerous SCIs listed in Table 1-1.

The "Door Control Unit" (DCU) Application, which is the main application, operates a door, detects and reports abnormal conditions. The DCU Boot Loader Application verifies the DCU hardware, and initializes the DCU to allow the DCU Application Update. The PTU Application allows software uploads to DCU, Door System data monitoring and Door System data log retrieval for troubleshooting. The DCU Neuron Gateway Application is responsible for the LonWorks network communications.

The SCM discipline is applied to all the Software Configuration Items (SCIs) identified for Door System Software project. The SCM implements the processes described within this SCM plan to ensure that products developed are correct, consistent, complete, and compliant with governing policies.

This plan addresses the SCM activities over the development and integration portion of the software product's life cycle. This SCMP applies to:

- Software being developed and delivered as part of the system;
- Software being developed and not delivered as part of the product (e.g. testware, tools);
- Software being developed and used for production purposes;
- Software re-used totally or in part;
- Programmable devices developed and delivered as part of the system;
- Commercially Available Software (CAS) incorporated into the product;
- CAS used as tools (e.g. compilers, word-processors, drawing tools);
- Customer-supplied software incorporated into the product;
- Customer-supplied software not incorporated into the product (e.g. testware, tools);
- Specifications;
- Interfaces;
- Development Environments;
- Test Environments;
- Media delivered containing any of the above;
- Documentation of all above.

Each Door System SCI is handled as individual item on the Bill of Materials (BOM) for the End Product. The Software Configuration & Serialization of systems being shipped are addressed by VSRS' Production Department. See Section 3.1.1.3.

Section 1.2.1.1 below provides information about the project structure and the SCIs. For further information on each of the software components, please refer to the SPMP document #201111207.

1.2.1 Software Configuration Items Covered

The Door System is composed of a number of subsystems. Table 1-1 lists the Software Configuration Identifications (SCIs) identified for the software system. Figure 1-1 identifies the SCI within each subsystem.

Table 1-1: SCI Nomenclature

VSRS P/N	SCI Nomenclature
2011L0053	DCU Boot Loader Application
2011L0051	DCU Application
2011L0059	DCU CPLD Binary File
2011L0052	Neuron Gateway Application
2011L0054	PTU Application

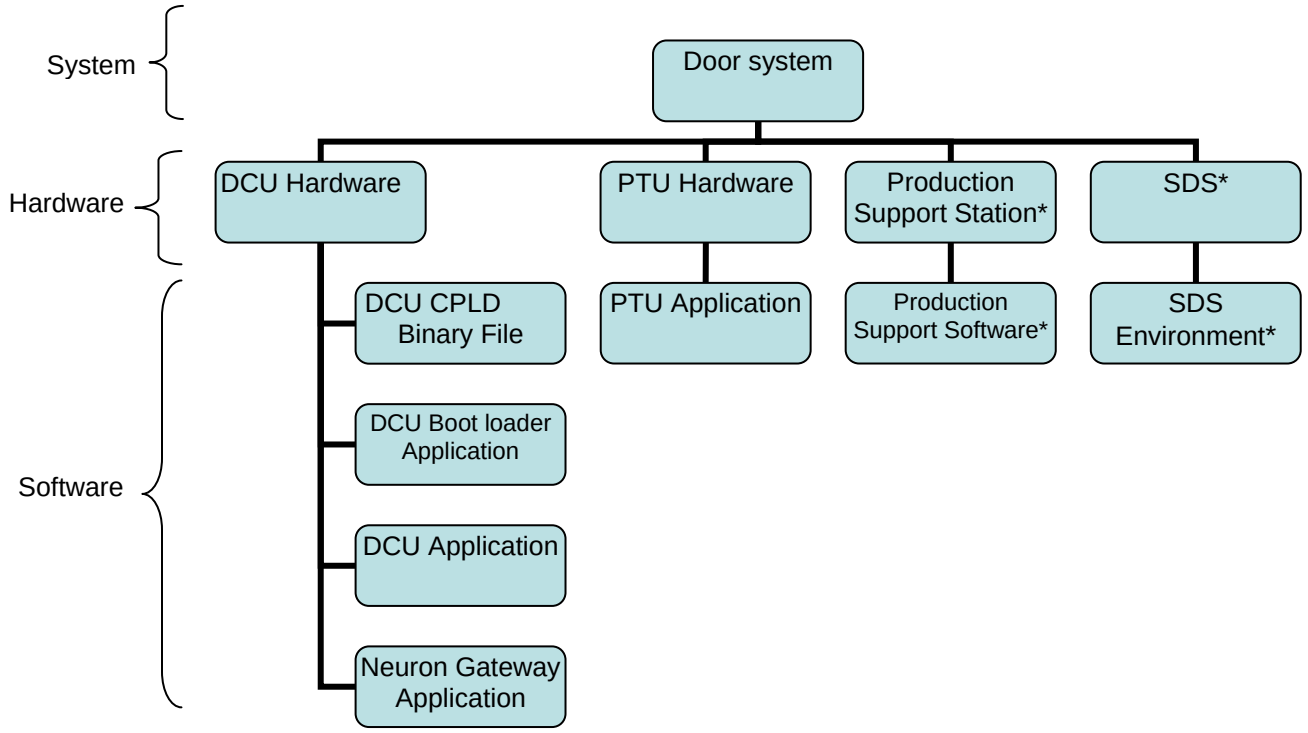

Figure 1-1: Project Structure

*These items are not SCI deliverables but they will be used internally

The Door System software is hosted on four hardware platforms: the Door Control Unit (DCU), the Portable Test Unit (PTU), the Production Support Station (PSS) and the Software Development Station (SDS).

- The DCU Hardware hosts the DCU Application, the DCU Boot Loader Application, the Neuron Gateway Application;
- The PTU Hardware hosts the PTU Application;
- The Production Support Station hosts the Production Support Software;
- The SDS hosts all the tools that are required to compile and maintain the above software.

VSRS has developed similar products. These products can be reused in some way. Having analyzed and estimated the technical aspects of the project and considering efficiency, stability of the software and the overall quality of the product, VSRS decided to adopt an approach of reusing or adjusting to what had been developed to be compliant to the Denver project requirements. For an overview description of the software components, please refer to SPMP document #201111207.

1.2.2 Software Configuration Management Relationships to Other Configuration Management

All departments working on the project shall comply with the ISO 9001 Standard and shall follow the procedures defined in the documents specified below:

- EWI003: Work Instruction for Project Folder, Structure, Naming, and Permissions;
- EWI006: Work Instruction for Document Control, File naming methodology;
- EWI008: Work Instruction for Drawing and Document Release to Public;
- EWI010: Work Instruction for ECD Process;
- 08-001: Work Instruction for Configuration Management Procedure.

1.2.3 Life Cycle

The SCM plan addresses the SCM activities over all phases of the software product's life cycle.

For additional information about the level of control applied throughout the software life cycle for configuration management, refer to Section 3.2.

1.3 Assumptions

It is assumed that throughout the life cycle of the project, the resources performing the SCM activities shall be trained and that there is sufficient disk space provided by IT in order to carry out the Configuration Management activities as planned.

1.4 Acronyms and Definitions

Definitions and acronyms listed below establish meaning in the context of this SCMP.

1.4.1 Acronyms

AB	Allocated Baseline
BOM	Bill of Materials
CAD	Computer-Aided Design
CAS	Commercially Available Software
CD	Compact Disk
CD-R	Compact Disk Recordable
CDRL	Contract Data Requirements List
CD-ROM	Compact Disk Read Only Memory
CM	Configuration Management (Manager)
CMM	Capability Maturity Model
CMMI	Capability Maturity Model Integration
COTSR	Commercial Off-The-Shelf Software Report
CPLD	Complex Programmable Logic Device
CPU	Central Processor Unit
DB	Developmental Baseline
DCG	Document Control Group
DOP	Door Operator
DTCR	Date and Time Compliance Report
ECD	Engineering Change Document
EPROM	Electrical Programmable Read Only Memory
EWI	Engineering Work Instruction
FB	Functional Baseline
FCA	Functional Configuration Audit
FMECA	Failure Modes, Effects and Criticality Analysis
FMI	Field Modification Instruction
FTA	Fault Tree Analysis
IAR	Improvement Action Request
IEEE	Institute of Electrical and Electronics Engineers
IO	Input / Output
IPD	Interface Protocol Document
ISO	International Standardization Organization
IT	Information Technology
ITP	Integration Test Plan
ITPS	Integration Test Procedure Specification
ITR	Integration Test Report
MONIT	Monitoring
MRP	Manufacturing Resource Planning
P/N	Part Number
PB	Product Baseline
PCA	Physical Configuration Audit
PDF	Portable Document Format
PID	Proportional Integral Derivative
PM	Project Management (Manager)
PRM	Project Review Meeting
PROM	Programmable Read Only Memory
PTA	Portable Test Application
PTS	Procurement Technical Specification
PTU	Portable Test Unit
QAM	Quality Assurance Manual
QSP	Quality System Procedure
RAMS	Reliability, Availability, Maintainability and Safety
RTD	Regional Transportation District
RTDM	Real Time Data Monitoring
RTP	Routine Test Procedure
SACS	Software Applications Configuration Specifications
SCCB	Software Configuration Control Board
SCI	Software Configuration Item
SCIST	Software Configuration Items Summary Table
SCM	Software Configuration Management (Manager)
SCMP	Software Configuration Management Plan
SCT	Software Coding and Testing
SCTR	Software Component Test Report
SD	Software Design
SDD	Software Design Description
SDL	Software Development Library
SDP	Software Development Plan
SDS	Software Development Station
SDS	Software Development Supervisor
SDT	Software Development Tracking
SEEM	Software and Electrical Engineering Manager
SEI	Software Engineering Institute
SEP	Software Engineering Process
SFD	System Functional Description
SPG	Software Plan Generation
SPI	Serial Peripheral Interface
SPMP	Software Project Management Plan
SPP	Software Project Planning
SPR	Software Problem Report
SPRM	Software Project Review Meeting
SQA	Software Quality Assurance (Agent)
SQAP	Software Quality Assurance Plan
SRD	Software Requirements Definition
SRM	Senior Review Meeting
SRS	Software Requirements Specification
SRTM	Software Requirements Traceability Matrix
SSHA	Sub-System Hazard Analysis
STL	Software Team Leader
SUM	Software User Manual
SV	Software Validation
SVD	Software Version Description
SVTP	Software Validation Test Plan
SVVP	Software Verification and Validation Plan
SVVR	Software Verification and Validation Report
SW	Software
SWE	Software Engineering (Engineer)
SWI	Software Work Instruction
SWSD	Software Workstation Support Document
TBD	To Be Determined
VSRS	Vapor Stone Rail Systems

1.4.2 Definitions

ABAK: Time Monitoring System (Project Tracking Database).

Allocated Baseline (AB): The initial approved specifications governing the development of configuration items that are part of a higher-level configuration item. The Allocated Baseline is performed at the authentication[1] of the Software Requirement Specifications. This Specification identifies the functional, interfaces and performance requirements of the Software Configuration Item (SCI).

Configuration Management (CM): A discipline applying technical and administrative direction and surveillance to: identify and document the functional and physical characteristics of a configuration item, control changes to those characteristics, record and report change processing and implementation status, and verify compliance with specified requirements.

Developmental Baseline (DB): These baselines are at the discretion of the project and shall serve to identify a version undergoing developmental and acceptance testing phases.

Developmental Configuration: The software and associated technical documentation that define the evolving configuration of a SCI during development. The developmental configuration is under the developer's control, and therefore is not called a baseline.

Functional Baseline (FB): The initial approved technical documentation for a configuration item. Functional Baseline is performed at the authentication of the System Specification or its equivalent (System Functional Description (SFD)). This Specification identifies the functional and performance requirements of the overall system under development.

Functional Configuration Audit (FCA): An evaluation conducted to verify that the development of a configuration item has been completed satisfactorily, that the item has achieved the performance and functional characteristics specified in the functional or allocated baseline, and that its operational and support documents are complete and satisfactory.

Physical Configuration Audit (PCA): An evaluation conducted to verify that a configuration item, as built, conforms to the technical documentation that defines it.

Product Baseline (PB): The initial approved technical documentation (including, for software, source code), defining a configuration item during the production, operation, and logistics support of its life cycle.

Quality Assurance: A planned and systematic set of all actions necessary to provide adequate confidence that the item or product conforms to the established technical and quality requirements.

Software Configuration Item (SCI): An aggregation of software that is designated for configuration management and treated as a single entity in the configuration management process.

[1] The first page must be scanned and incorporated in the document as the first page to prove authenticity.

1.5 References

This section lists the number, title, revision, date, and source of all documents referenced in this SCMP.

Standards

CMU/SEI-2006-TR-008	CMMI® for Development, Version 1.2, August 2006, Software Engineering Institute, Carnegie Mellon University, www.sei.cmu.edu
IEEE Std 828-1998	IEEE Standard for Software Configuration Management Plans, June 25, 1998, IEEE, www.ieee.org
IEEE Std 1558-2004	IEEE Standard for Software Documentation for Rail Equipment and Systems, March 23, 2005, IEEE, www.ieee.org
BS EN50128	Railway Applications – Communications, Signaling and Processing Systems – Software for Railway Control and Protection Systems

VSRS Documents

CMM-01	Software Engineering Process, Rev. 3, 2001-03-31
CMMI-02	Software Engineering Policy Manual, Rev. 1, 2008-07-18
EWI-003	Project Folder, Structure, Naming and Permissions, Rev. 2, 2005-09-14
EWI-006	Document Control, File Naming Methodology, Rev. 4, 2005-09-14
EWI-008	Drawing and Document Release to Public, Rev. A, 1999-09-28
EWI-010	Engineering Change Document (ECD) [Process], Rev. B, 2005-09-14
QSP-04	Design Control, Rev. 5.5, 2005-07-19
QSP-S14-02	Data Storage, Preservation and Security, Rev. 3.0, 2006-08-4
SWI-01	Software Configuration Management (SCM) Guide, Rev. 2, 2001-03-27
SWI-02	Software Coding Standard, Rev. 3, 2001-03-27
SWI-05	Inspection Guide, Rev. 4, 2003-02-20
SWI-13	Software Quality Assurance (SQA) Guide, Rev. 3.0, 2000-08-23
S01-01	Software Problem Report Form, Rev. 21, Date Unknown
S01-02	Procedure to produce a CD-R Disk (Template), Rev. 3, 2001-09-12
S01-03	Procedure to produce a Floppy Disk (Template), Rev. 1, 2000-12-12
S13-02	Improvement Action Request Form, Rev. 1.1, 2001-01-30
S16-01	Risk Form, Rev 2.1
S17-04	Software Configuration Management Plan (Template), Rev. 2, 2003-01
04-WI-04	Work Instruction for Identification & Revision Control Procedure, Rev. 9, 2005-10-31
04-002	Engineering Change Document Form, Rev. 7, 2006-11
08-WI-001	Work Instruction for Configuration Management Procedure, Rev. 4.0, August 2001
08-002	Certificate of Compliance Form, Rev. 0, not dated
08-003	Serial Number, Revision, Modification Control Form, Rev. 2, May 2010
201111207	Software Project Management Plan, Door System, Rev. 0, 2011-06-03

201111208	Software Quality Assurance Plan, Door System, Rev. 1, 2011-05-02
201111254	Software Verification and Validation Plan, Door System – DCU Application, Rev. 0, 2011-06-10

Hyundai Rotem Documents

REDE112784	PTS Entrance Door for Denver Eagle P3 Project, Rev. 3, 2011-01-13

RTD Denver Documents

RFP No. 18FH012	RTD FasTracks Request for Proposal, 2009-09-30

2 SCM MANAGEMENT

This section describes the organization, tasks, and responsibilities of the project, with special emphasis on software configuration management.

2.1 Organization

Project organization chart and role description is available in the SPMP document #201111207.

2.2 SCM Responsibilities

2.2.1 SCM Roles

Table 2-1 identifies the specific project's organizational elements responsibilities against the activities of Section 3. For details on the overall software roles and responsibilities, refer to the SPMP document #201111207.

Table 2-1: SCM Responsibilities

Activity as per Section 3	Responsibility
3.1 Configuration Identification	
3.1.1 Configuration Item Identification	STL & SDS
3.1.2 Configuration Item Naming	STL & SDS
3.1.3 Acquiring Configuration Items	STL
3.2 Configuration Control	
3.2.1 Requesting changes	Customer through PM, STL, and SWE/Software (SW) Designers
3.2.2 Evaluating changes	STL / SW Engineers & Designers
3.2.3 Approving Changes	SCCB
3.2.4 Implementing changes	SWE / SW Engineers & Designers
3.3 Configuration Status Accounting	SCM
3.4 Configuration Audits and Reviews	SQA
3.5 Interface Control	STL
3.6 Subcontractor and Vendor Control	Not applicable

2.2.1.1 Software Configuration Control Board

The following subsections provide an overview of the functions, responsibilities, and authority of the SCCB.

2.2.1.1.1 Purpose and Objectives

The SCCB is composed of technical and administrative representatives who recommend approval or disapproval of proposed engineering changes to an SCI and its documentation. The SCCB also participates in the product baseline establishment and release.

2.2.1.1.2 Membership and Affiliation

The Software Configuration Control Board (SCCB) is a group made of project members. Its members are listed in Table 2-2. The Software Team Leader leads the SCCB.

2.2.1.1.3 Period of Effectiveness

The SCCB begins approving changes to an SCI and its documentation after the first product baseline established and approved.

2.2.1.1.4 Scope of Authority

The SCCB approves the software changes. It is the responsibility of the SQA to close the Software Problems Reports (SPRs) after reviewing the test results and supporting documentation according to SCCB's directives.

Issues that the project's SCCB is unable to resolve or involves a change in scheduling or fiscal costs are initially addressed by the SCCB, and then addressed at a Project Review Meeting (PRM) to follow up or Senior Review Meeting (SRM) for final approval or disapproval and recommendations.

2.2.1.1.5 Operational Procedures

For information on SCM activities performed by review board or a group, please refer to SWI-01 and 08-WI-001.

Table 2-2: SCCB Members

SCCB Member	Role
Martin Ouellet	SDS/SCM
James Kim	STL
Thomas Marmeth	Software Designer
Jose Trejo	Software Tester
Slavica Lezaic	SQA
Luc Bessette	Assessor
David Beaudoin	SEEM
Jocelyn Arcand	RAMS Engineer
Eric Valin	Lead Electrical Engineer
Pascal Parent	Lead Mechanical Engineer
Luc Giguère	Leader Engineering Projects
Aldo Di Rocco	Project Manager

2.3 Applicable External Plan Policies, Directives, and Procedures

This SCMP shall comply with the format and contents requirements of IEEE Std. 828-1998, as expressed in Section 6 of the Standard, EN50128, and IEEE Std. 1558-2004.

The following is a synopsis of the applicable policies, directives, and procedures referenced in this SCMP:

1. CMM-01 (Software Engineering Process): The Software Engineering Process (SEP) is a framework for the development of software projects at VSRS. The SEP follows rules, requirements and activities required in the Capability Maturity Model (CMM) for Software, level 2, from the Software Engineering Institute (SEI).

   The SEP establishes a common process framework that defines, implements, and supports the VSRS organization by specifying the methods, people and tools necessary to achieve projects and software products with the lowest cost, best delivery time and the lowest rates of defect.

   The SEP delineates the framework that shall be applied in any research, design, development, maintenance or acquisition of software products by VSRS. The SEP provides a tool for VSRS management to ensure that the people, products and processes involved, are supported by an organization and a set of methods and tools capable of meeting the standards of performance required by company objectives.

VSRS personnel must apply a predictable engineering process to design, develop, subcontract or maintain software products.

2. CMMI-02 (Software Engineering Policy Manual): VSRS is committed to use the SEP, CMM-01 and its related guides, forms and templates to develop, subcontract and maintain all new software projects or activities to achieve the quality, costs and schedule objectives of a project containing software.

   VSRS is committed to use the software practices, methods, and tools described in the SEP to meet the requirements management, software project planning, software project tracking and oversight, software subcontract, software quality assurance, and the software configuration management activities.

   VSRS is committed to use the best practices recommended by CMMI for software development.

   VSRS is committed to develop and continuously improve the SEP process to achieve the following objectives:

   - Ensure quality products and services that meet requirements.
   - Achieve technical compliance of delivered products and services.
   - Minimize negative impact on schedule and cost.
   - Maintain continuity during the different phases of design, development and maintenance contracts.
   - Meet the CMMI Level 2 requirements.
3. SWI-01 (Software Configuration Management Guide): This document describes the Software Configuration Management process used by VSRS to define and identify software baselines along with their components and documentation, to control changes to those baselines, to verify the conformance of those baselines to the original documentation and/or to the documented changes made to them and to maintain records of the baselines status. This guide defines outlines to all users and ensures that different individuals can repeat a set of auditable activities with appropriate diligence.
4. SWI-02 (Software Coding Standard): This document describes the coding standard for software programs. The purpose of this coding standard is to improve maintainability, re-usability, and source-code quality.
5. SWI-05 (Inspection Guide): This document describes the methods, resources, and the process to be used by VSRS to perform an Inspection to remove defects on a software document early and efficiently during a software project. This guide defines outlines to all users and ensures that a set of auditable activities can be repeated by different individuals with appropriate diligence.
6. SWI-13 (Software Quality Assurance Guide): This document describes the methods, resources, and processes to be used by VSRS to conduct and perform Software Quality Audit, Software Review and Configuration Audit to evaluate the status, the adequacy, and the effectiveness of the Quality Software activities conducted by VSRS. This guide defines outlines to all users and ensures that a set of auditable activities can be repeated by different individuals with appropriate diligence.
7. EWI-003 (Project Folder, Structure, Naming, and Permissions): This document describes a project folder structure, naming methodology, and security access permissions for the Engineering folder.
8. EWI-006 (Document Control, File naming methodology): This document describes a mandatory file naming methodology to uniquely identify a specific file and the revision level of the respective file. This work instruction also explains how documents and drawings must be identified or be referenced to other documents.
9. EWI-008 (Drawing and Document Release to Public): This document describes the steps to release the drawings and production related documents to public.
10. EWI-010 (Engineering Change Document): This document describes the steps for initiating, completing and approving all Engineering Change Documents (ECD).
11. QSP-04 (Design Control): This document describes the method to develop and implement design strategies which will enable the company to satisfy the needs and expectations of its customer. It also describes the method to develop and operate the design function such that it is fully integrated with the other company functions, which contribute to the successful completion of the product design.

12. QSP-S14-02 (Data Storage, Preservation and Security): This document describes the methods to be used by VSRS to assure that all electronic documents are backed-up, maintained and preserved.
13. S01-02 (Template for Procedure to produce CD-ROM) and S01-03 (Template for Procedure to produce Floppy Disk): These documents describe the method to produce and verify the construction integrity of a floppy disk (S01-03) and a CD-ROM (S01-02).
14. S13-02 (Improvement Action Request Form): This document indicates the non-conformance, standard requirements that shall be followed, the immediate improvement required, and the root cause of the non-conformance, and the permanent improvement to be implemented to eliminate the recurrence of that type of non-conformance.
15. S17-04 (Template for SCMP): This document is the SCMP Template, which defines the format and the contents of the SCMP.
16. 04-WI-04 (Work Instruction for Identification & Revision Control Procedure): This document describes the process of initiating, implementing and recording FMI's on delivered products. The process of determining and setting Modification levels is also described. The identification of Assembly Levels is also reviewed.
17. 04-002 (Engineering Change Document Form): This Form is used for ECD Process. Refer to Work Instruction #EWI-010.
18. 08-WI-001 (Work Instruction for Configuration Management Procedure): This document defines the method by which Serialization and Configuration Control will be used on specified items (unit, device or portion) and/or Software.
19. 08-002 (Certificate of Compliance Form): This Form is used to certify that the attached items and quantities shown on the Sales Order are inspected and or tested to approve drawings. Refer to Work Instruction #08-001.
20. 08-003 (Serial Number, Revision, and Modification Control Form): This form shall be completed each working week when serial numbers are issued and passed Quality Assurance Configuration. Refer to Work Instruction #08-001.

Upon changes to any of the following documents, the SCM shall assess the impact on this SCMP and submit change requests to the SCCB:

- The VSRS' CMMI-02 (Software Engineering Policy Manual) applies, and specifically the Software Configuration Management policies.
- The SCM activities and practices are governed by the CMM-01 (Software Engineering Process) and more specifically by SWI-01 (SCM Guide).
- This SCMP is developed according to the directives of the S17-04 (SCMP Template), IEEE 828-1998 and IEEE 1558-2004.

2.4 SCM Risk and Abatement

All risks that may have an impact on the implementation of SCM activities shall be raised by the SCM, the Team Leader or the SQA. The mitigation and contingency plans related to the SCM risks are discussed with all the team members during the SPRM.

2.4.1 Risk Identification

The SCM risks are identified in a Software Risk Evaluation Form along with other risks associated to the project. The risk form contains the following information for each risk:

- A description of the risk,
- Risk type,
- Risk category,
- Risk level,
- Probability,
- Impact,
- Exposure,
- Mitigation plan,
- Input criteria,
- Contingency plan.

2.4.2 Risk Management

The SCM risks identified on the Software Risk Evaluation Form are reviewed during each SPRM and the Software Risk Evaluation Form is updated accordingly. After each update, the Software Risk Evaluation Form is distributed to the stakeholders for their signature so that the stakeholders are aware of the risks and can intervene to reduce or eliminate the risks.

3 SCM ACTIVITIES

This section identifies all functions and tasks required when managing the configuration of the software system. Figure 3-1 provides an overview of the SCM Functions.

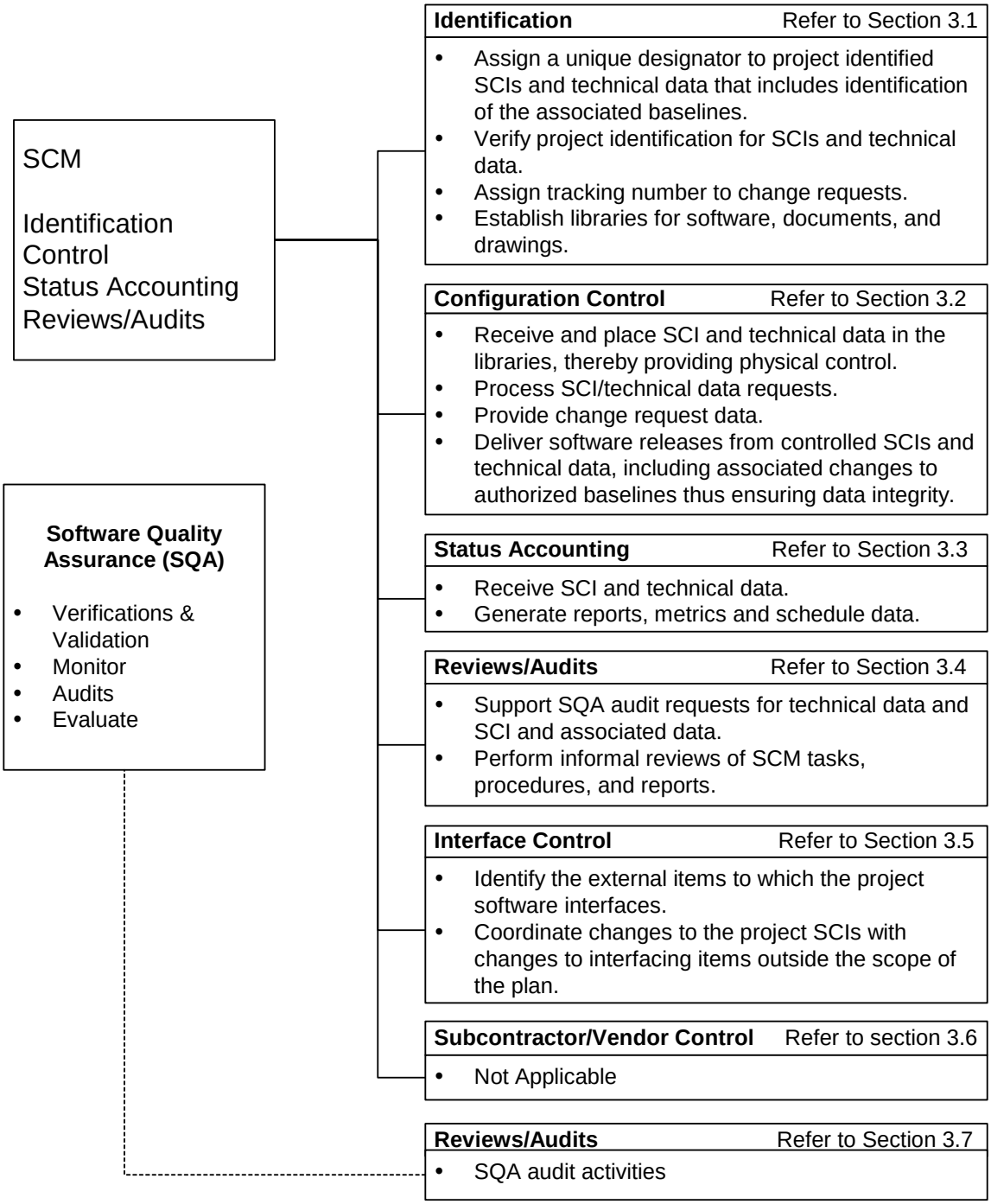

Figure 3-1: Overview of the SCM Functions

Details of activities, responsibilities, and dependencies of the resources are presented in the VSRS' *CMM-01: Software Engineering Process*.

This section covers the following life-cycle phases of the CMM-01's Software Development Process module:

SPP-200	Software Project Planning
SPG-300	Software Plan Generation
SRD-400	Software Requirements Definition
SD-500	Software Design
SCT-600	Software Coding and Testing
SV-700	Software Validation
SDT-800	Software Development Tracking

3.1 Configuration Identification

This section identifies how each configuration item and its versions are to be uniquely named and describes the activities performed to define, track, store, and retrieve configuration items. The project configuration items and their structures at each project control point are identified.

3.1.1 Configuration Item Identification

This section describes the process of configuration identification.

3.1.1.1 Configuration Items

The selection of SCIs is the project management or the developer's responsibility. SCIs are placed under configuration management in accordance with this plan. Once the SCI has been identified and provided to the SCM, the SCMP will be updated.

3.1.1.2 Baseline Definition

For each SCI, configuration identification is established for software technical documentation, source code, executable, media, and environment. The initially approved configuration identification establishes baselines from which subsequent changes are controlled. The configuration identifications and baselines for the SCIs are defined as shown below (see Section 1.4.2 for detailed definitions of the baselines and the events that create these baselines).

- **Functional Baseline (FB):** Marked in column FB of Table 3-1 are the documents that comprise the Functional Baseline.
- **Allocated Baseline (AB):** Marked in column AB of Table 3-1 are the documents that comprise the Allocated Baseline.
- **Developmental Baseline (DB):** Marked in column DB of Table 3-1 are the documents that comprise the Developmental Baseline.
- **Product Baseline (PB):** Marked in column PB of Table 3-1 are the documents that comprise the Product Baseline.

A baseline is established once all documents applicable to a baseline (as per Table 3-1) are under configuration control.

The subsequent changes to a baseline are controlled as per Section 3.2.

Please refer to SWI-01 (Software Configuration Management Guide) for procedures used to establish software baselines and to control changes to those baselines.

Table 3-1: Baseline Establishment

Configuration Item	FB	AB	DB[1]	PB
RTD Denver EMU Door System				
Technical description	X			
Hyundai Rotem Technical Documentation				
Procurement Technical Specification	X			
System Documentation				
System Functional Description (SFD)	X			
Software Requirements Traceability Matrix (SRTM), including PTU App., 1 per System.		X		
Software Configuration Items Summary Table (SCIST)	X			
Software Workstation Support Document (SWSD)				X
Commercial Off-the-Shelf Software Report (COTSR)				X
Date and Time Compliance Report (DTCR)				X
System Test Plan (SyTP)***		X		
Software Plans and Reports				
Software Project Management Plan (SPMP)		X		
Software Quality Assurance Plan (SQAP)		X		
Software Configuration Management Plan (SCMP)		X		
Software Validation and Verification Plan (SVVP)		X		
Software Validation and Verification Report (SVVR)				X
Software Acceptance Test Report (SATR)			X	X
Software Version Description (SVD)			X	X
Software Applications Configuration Specifications (SACS)		X		X
Door Control Unit Hardware				
DCU Application				
Software Requirements Specification (SRS)		X		
Master Software Design Description (SDD)				X
Module Software Design Description (SDD)				X
Software Component Test Plan (SCTP)*		X		
Software Integration Test Plan(SITP)**		X		
Software Integration Test Report (SITR)			X	X
Software Component Test Report (SCTR)			X	X

[1] This baseline can be used whenever the testing phase is not completed by the customer or when an SCI full product baseline release can delay the car builder or customer testing.

*SCTP will be created for the DCU Application and includes 3 major sections: Software Component Test Planning, Software Component Test, Design Specification, and Software Component Test Procedures Specification. For further information check SQAP chapter 4.9.

**SITP will be created for each of the SCIs and includes 3 major sections: Software Integration Test Planning, Software Integration Test Design Specification, and Software Integration Test Procedures Specification. For further information check SQAP chapter 4.9.

***SyTP includes 3 major sections: System Test Planning, System Test Design Specification, and System Test Procedures Specification. For further information check SQAP chapter 4.9.

Configuration Item	FB	AB	DB[1]	PB
Source Code				X
Executable			X	X
Development Environment				
Development Tools				X
CD-ROM Media with sources code				X
CD-ROM Media with executable				X
DCU Boot Loader Application				
SRS		X		
SDD				X
SITP**		X		
SITR			X	X
Source Code				X
Executable			X	X
Development Environment				
Development Tools				X
CD-ROM Media with sources code				X
CD-ROM Media with executable				X
DCU Neuron Gateway Application				
ICD		X		
SRS		X		
SDD				X
SITP**		X		
SITR			X	X
Source Code				X
Executable			X	X
Development Environment				
Development Tools				X
CD-ROM Media with sources code				X
CD-ROM Media with executable				X
Portable Test Unit Hardware				
PTU Application				
ICD		X		
SRS		X		
SDD				X
SITP**		X		
SITR			X	X
SUM				X
Source Code				X
Executable			X	X

Configuration Item	FB	AB	DB[1]	PB
Development Environment				
Development Tools				X
CD-ROM Media with sources code				X
CD-ROM Media with executable				X
Production Support Station				
Production Support Software				
CD-ROM Media with executable				X
DCU Software Programming Procedure				X
Software Development Station				
Software Development Environment				
CD-ROM Media with executable				X

3.1.1.3 Engineering Release

The software release process begins at the establishment of the system integration and testing. Prior to this release process, software Product Baseline shall be approved by the project's SCCB. The software release that is identified as part of the PB is provided for integration with the operational hardware. Final acceptance testing is performed by both the Car-Builder and the transit authority. Issues found are resolved using the baseline change process through SPR forms and ECD. Upon satisfactory completion, the software release is approved as the Product Baseline Configuration. Approval for service use initiates distribution to fleet users, and production.

3.1.2 Configuration Item Naming

The following sections describe the methods used in identifying the SCI and associated technical data and project-developed support software required for the development and test.

As stated in Section 1.2, each Door System SCI is handled as individual item on the BOM for the End Product. The Software Configuration & Serialization of systems being shipped are addressed by VSRS' Production Department as per 08-WI-001 (Work Instruction for Configuration Management Procedure), 04-WI-04 (Work Instruction for Identification & Revision Control Procedure), EWI-006 (Work Instruction for Document Control, File Naming Methodology), Form 08-002 (Certificate of Compliance), and Form 08-003 (Serial Number, Revision, Modification Control Form). Each Work Instruction and Procedure listed is part of the VSRS Quality Assurance Manual (QAM). The verification process is managed by VSRS' Quality Department.

The Table 3-2 provides, for each SCI, the related VSRS Part Number (P/N). More details on VSRS documentation numbering are provided in the following sub-sections.

3.1.2.1 Documentation

The method of identification of each item will be as per EWI-006. Each page of the document contains the identification number with the applicable revision letter/number.

Documents not yet baselined have assigned a one or two digits revision number, starting from zero (e.g. Rev 0, Rev. 10). Baselined documents have assigned a one or two revision letter, starting from A (e.g. Rev. A, Rev. AA).

3.1.2.2 Source Code

Each SCI and all project-developed support software required for development is identified with unique names, acronyms, and numbers. SCIs have an assigned unique identifier composed of the subsystem's or sub-assembly's name (e.g. DCU), followed by one or more words (e.g. DCU Boot Loader Application, PTU Application).

Software units (source files) are identified by the developer, as see fit. The version of each individual file has an assigned unique identifier by the SCM tool (Visual SourceSafe).

Software versions not yet baselined are numbered in the form *Major.MinorX* where X is an incremental letter during the development of the software (e.g. v1.2a → v1.2b). Baselined software versions are numbered in the form *Major.Minor* (e.g. v1.1 → v1.2). The version starts at 1.0. The *Major* part is incremented and the *Minor* part is set to zero if the version is not form-fit-function compatible with its predecessor (e.g. v1.2 → v2.0). If the version is form-fit-function compatible with its predecessor, only the *Minor* part is incremented (e.g. v1.2 → v1.3). The Major and Minor are each composed of integer numbers from 1 to 3 digits long (e.g. v2.102).

Release of software is performed by mean of releasing its by-products. Executable files package, source files package and media are examples of by-products. Before a software version is released, the SCM applies a version label to the software using the SCM tool's *label* feature. The version label is composed of the SCI identifier, and the software version. The naming methodology used for releasing the source code files is defined in EWI-006.

Note: Not all versions make their way into a release. The Software Team Leader is responsible to identify which software versions are internal baselined and which software versions are baselined to be released. Baseline process is documented in the SCM Guide document #SWI-01 as stated in Section 2.3.

3.1.2.3 Executable Code

The executable files for each SCI are identified by the developer. Each version of the executable file has a unique identifier assigned by the SCM tool (Visual SourceSafe).

Software versions not yet baselined are numbered in the form *Major.MinorX* where X is an incremental letter during the development of the software (e.g. v1.2a → v1.2b). Baselined software versions are numbered in the form *Major.Minor* (e.g. v1.1 → v1.2). The version starts at 1.0. The *Major* part is incremented and the *Minor* part is set to zero if the version is not form-fit-function compatible with its predecessor (e.g. v1.2 → v2.0). If the version is form-fit-function compatible with its predecessor, only the *Minor* part is incremented (e.g. v1.2 → v1.3). The Major and Minor are each composed of integer numbers from 1 to 3 digits long (e.g. v2.102).

The naming methodology used when releasing the executable files is defined in EWI-006.

3.1.2.4 Media

Software Media are:

- Compact Disk Recordable (CD-R);
- Floppy Disks;
- Re-programmable and program-once-only devices (e.g. Programmable Read Only Memory (PROM), Electrical Programmable Read Only Memory (EPROM), CPLDs).

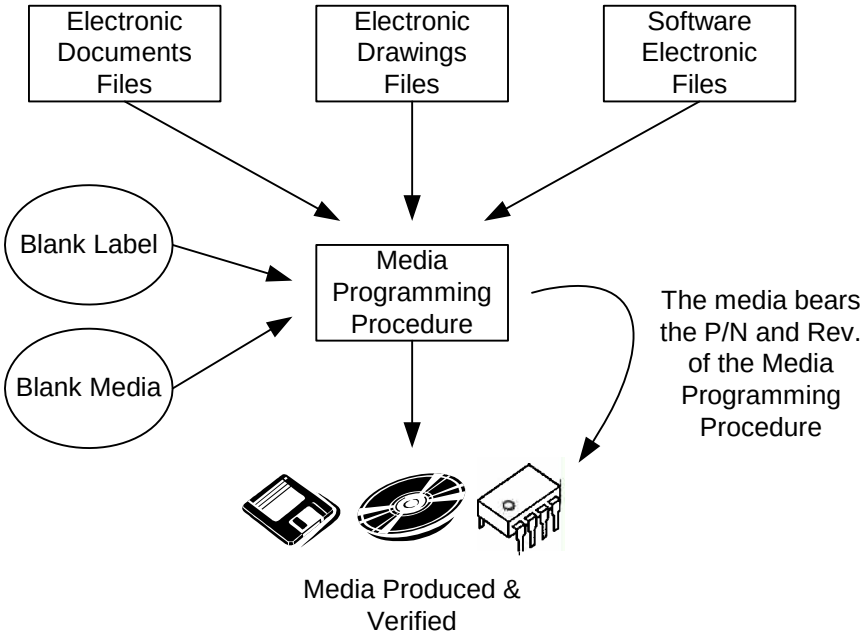

Figure 3-2: Media Identification

- **Identification:** The method of identification of media will be as per SWI-01, S01-02 (floppy disk), S01-03 (CD-R) and EWI-006 (numbering). The method is summarized below and represented in Figure 3-2.
- **Revisions:** Media not yet baselined have an assigned one or two digits revision number, starting from zero (e.g. Rev. 0, Rev. 10). Baselined media have assigned a one or two revision letter, starting from A (e.g. Rev. A, Rev. AA).
- **Label & Marking:** Each media volume contains an identification number as per EWI-006, the volume number, the applicable revision letter/number of the media, and the title/description of the content. The identification number denotes a document (as per the S01-02 or S01-03 template) containing the software, hardware, material requirements, and the procedure necessary to create and verify the media set (media, files that it contains, and the labels).

3.1.2.5 Commercial Software Identification

Vendor supplied media are repackaged into a VSRS' P/N and identified as per Sections 3.1.2.4 and 3.1.2.6.2.

3.1.2.6 Other Configuration Item Identification

3.1.2.6.1 Drawing Identification

The method of identification of each item will be as per EWI-006. Each page of the drawing contains the identification number with the applicable revision letter/number.

Drawings not yet baselined have an assigned one or two digits revision number, starting from zero (e.g. Rev 1, Rev. 10). Baselined drawings have an assigned one or two revision letter, starting from A (e.g. Rev. A, Rev. AA).

3.1.2.6.2 Software Electronic Items

Software electronic items are files that do not have (yet) a physical media (see also Section 3.1.2.4). In the method described in EWI-006, software electronic items are referred as “L” types (e.g. 2001L0001-00R-A).

The method of identification of each item will be as per EWI-006.

Items not yet baselined have an assigned one or two digits revision number, starting from zero (e.g. Rev.1, Rev. 10). Baselined items have an assigned one or two revision letter, starting from A (e.g. Rev. A, Rev. AA).

3.1.2.6.3 Identification of Customer Supplied Media

Customer supplied media are repackaged into a VSRS’ P/N and identified as per Sections 3.1.2.4 and 3.1.2.6.2.

3.1.2.6.4 Identification of Test Report/Records

Test Report/Records results are hand written on the test procedures. The reports are then packaged into a VSRS’ P/N and identified as per Sections 3.1.2.4 and 3.1.2.6.2.

3.1.2.6.5 SCIs with the Physical Marking on the Top Drawing

VSRS has a mechanism in place to identify the hardware and software modification level (mod level) on the exterior of the components required for the Door System. On the top level drawing, one can always find the mod (modification) level from the Physical Marking and the information about all the SCIs required in the system.

When a part is marked with a mod level other than that on the latest top level drawing, it needs to be upgraded. This upgrade may be either hardware or software related (hardware upgrade is out of scope of the document). In the case of software upgrade, there is always Field Modification Instructions (FMIs) documented to describe how to upgrade the software and to which new mod level this upgrade is to apply. As a result of the mod level change, the top level drawing should also be updated and released.

To guarantee that the software is upgraded properly, one must strictly follow the FMI. Note that there are circumstances where he/she should follow more than one FMI in order to upgrade the particular software program.

In summary, VSRS keeps track of any SCI related changes/upgrades by having following artifacts:

- Upgraded software package,
- FMI document,
- Updated top drawing, and
- Updated SVD.

Table 3-2: Configuration Items Deliverables

Configuration Item	VSRS P/N	CDRL
Safety Related Deliverables		
Sub-System Hazard Analysis (SSHA)	201111368	11
Fault Tree Analysis (FTA)	201111188	17
Failure Mode Effects and Criticality Analysis (FMECA)	201111189	16
System Deliverables		
Software Project Management Plan (SPMP)	201111207	22
Software Quality Assurance Plan (SQAP)	201111208	23

Configuration Item	VSRS P/N	CDRL
Software Configuration Management Plan (SCMP)	201111186	30
Software Validation and Verification Plan (SVVP)	201111217	28
System Functional Description (SFD)	201111185	25
Software Verification and Validation Report (SVVR)	201111446	29
Software Requirements Traceability Matrix (SRTM)	201111448	34
Software Version Description (SVD)	201111449	38
Software Applications Configuration Specifications (SACS)	201111576	148
DCU Boot Loader Application Deliverables		
Software Requirements Specifications (SRS)	201111271	144
Software Design Description (SDD)	201111456	161
Software Integration Test Plan (SITP[1])	201111580	166
Software Integration Test Report (SITR)	201111458	
Executable – Boot Loader Application	2011L0053	
DCU Application Deliverables		
Software Requirements Specifications (SRS)	201111270	145
Software Design Description (SDD) Documents	-	-
Master SDD – DCU Application	201111450	159
Module SDD – IO Module	201111628	158
Module SDD – MONIT Module	201111630	160
Module SDD – Serial Driver Module	201111582	156
Module SDD – Watchdog Module	201111544	157
Module SDD – DOP Module	201111568	149
Module SDD – Fault Logger Module	201111583	150
Module SDD – PID Module	201111554	151
Module SDD – PTA Protocol Module	201111586	152
Module SDD – Real-Time Clock Module	201111581	153
Module SDD – RTDM Module	201111585	154
Module SDD – Self-Test Module	201111584	155
Module SDD – SPI Module	201111659	
Software Component Test Plan (SCTP[2])	201111451	170
Software Integration Test Plan (SITP[1])	201111453	167
Software Integration Test Report (SITR)	201111454	
Executable – DCU Application	2011L0051	
DCU Neuron Gateway Application Deliverables		
Software Requirements Specifications (SRS)	201111256	146
Software Design Description (SDD)	201111459	
Software Integration Test Plan (SITP[2])	201111460	168
Software Integration Test Report (SITR)	201111461	
Interface Control Document (ICD) - SPI	201111628	163
Executable -- Neuron Gateway Application	2011L0052	
PTU Application Deliverables		
Software Requirements Specifications (SRS)	201111462	147
Software Design Description (SDD)	201111463	
Software Integration Test Plan (SITP[1])	201111464	169
Software Integration Test Report (SITR)	201111465	
Software User Manual (SUM) – PTU	201111466	165
Interface Control Document (ICD) – Portable Test	201111489	164
Executable -- PTU Application	2011L0054	

[1] SITP will be created for each of the SCIs and includes 3 major sections: Software Integration Test Planning, Software Integration Test Design Specification, and Software Integration Test Procedures Specification. For further information check SQAP chapter.4.9

[2] SCTP will be created for DCU Application and includes 3 major sections: Software Component Test Planning, Software Component Test, Design Specification, and Software Component Test Procedures Specification. For further information check SQAP chapter. 4.9

3.1.3 Acquiring Configuration Items

The following libraries are established:

- Software Development Library (SDL);
- Drawing Library;
- Software Tools Library;
- MRP Library.

This project does not keep any physical media (magnetic or paper) under configuration management. Rather, the data and documentation are maintained in electronic form in one of the libraries listed above. The software baselines are defined and established according to the SWI-01 and the data of each baseline is stored in the Software Development Library and the Drawing Library. By using the information defined in the SVD (i.e. the list of documents and their configuration), the controlled items for a specific baseline can be retrieved and reproduced from the Software Development Library and the Drawing Library.

The baselined item is also stored in the MRP Library prior of being delivered to the customer.

SWI-01 (Software Configuration Management Guide): This document describes the Software Configuration Management process used by VSRS to define and identify software baselines.

3.1.3.1 Documentation

Locations: \\vcisrv002\projects\7230 - Denver EMU Eagle P3 Door System\Engineering Released\Software\

All released software documentation related to the project is placed under the controlled Software Development Library (SDL). The SDL provides storage of and controlled access to software documentation in human-readable form. The SDL components are initially documented in an identification list for the Allocated Baseline. Section 3.1.1 describes identification for the SDL collection. Throughout the life cycle of the project, the software documents are validated by SQA and verified by SCM. SCM uses the project reviews and the Inspection Process to perform these verifications. SCM verifies that approved software changes have been incorporated into the proper location of the SDL, reports status to the SCCB, and performs the release function upon SCCB authorization.

When releasing the generic software document to a customer, the copy of it is stored in the location specified above. The CDRL number is stamped in PDF if it is applicable.

3.1.3.2 Source code

Locations: \\vcisrv002\ProjectsDB\VSS\7230-Denver

Tool: Visual SourceSafe

Format: Visual SourceSafe

The Software Development Library (SDL) is the controlled collection of intermediate source code. The SDL provides storage of and controlled access to software development products in human-readable form, machine-readable form, or both.

The SDL consists of a series of revisions corresponding to the different phases through which the software is developed. Throughout the life cycle of the project, the software products are validated by SQA and verified by SCM. SCM uses the project reviews and the Code Walkthrough Process to perform these verifications. SCM verifies that approved software changes have been incorporated into the proper location of the SDL, reports status to the SCCB, and performs the release function upon SCCB authorization.

Source code shall comply with the requirements of SWI-02.

3.1.3.3 Library and Component Code

3.1.3.3.1 Drawing Library

Location: \\vcisrv002\projects\7230 - Denver EMU Eagle P3 Door System \Engineering\Mechanical+Electrical\Drawings

Tool: None

Format: File System

The Drawing Library provides storage of and controlled access to drawings in Computer-Aided Design (CAD) readable form, human-readable form, or both. The Drawing Library components are initially documented in an identification list for the AB. The Section 3.1.1 describes identification for the Drawing Library collection.

3.1.3.3.2 Software Tools Library

Location: \\vcisrv002\projects\7230 - Denver EMU Eagle P3 Door System\Engineering\Software\Tool

Tool: None

Format: File System

The Software Tools Library is the controlled collection of compilers, off-the-shelf products, and operating systems that comprise the development environment, test environment, and production environment of the SCIs. The Software Tools Library provides storage of and controlled access to software tools in a form suitable for network installation of the tool, for repackaging of the tool onto a media, or both. The Software Tools Library components are initially documented in an identification list for the Product Baseline. The Section 3.1.1 describes identification for the Software Tools Library collection.

For each tool, the following information should be provided:

- Installation and configuration procedure of the tool in the context for which it is to be used;
- Legal statements (kind of license, number of licenses, ownership);
- License numbers, Compact Disk (CD)-Keys;
- Source of the tool (if downloaded) or manufacturer, point of contact;
- Related tools (e.g. service packs, additional components);
- Special requirements (e.g. access rights, other software already installed).

3.1.3.4 Physical Storage

Libraries are maintained onto VSRS' corporate server vcisrv002. Project Folder, Structure, Naming, and Permissions are governed by EWI-003 and managed by VSRS' Management Information Service group (outside the scope of this SCMP). The accesses to the file network server are controlled by a username and a password allocated to the personal on a project and a functional basis (i.e. Marketing, Engineering, Finance, etc.). Backup of the servers is governed by QSPs-14-02 and managed by VSRS' Management Information Service group (outside the scope of this SCMP). Each Work Instruction and Procedure referenced above is part of the VSRS Stone Rail System QAM.

3.1.3.5 Configuration Item Retention and Recovery

The retention period for all libraries is 10 years. The procedures for data recovery are managed by VSRS' Management Information Service group and are outside the scope of this SCMP. For information regarding the data recovery process, refer to document QSPs-14-02.

3.1.3.6 Configuration Item Distribution

Location: http://www.vri.wabtec.com/

Tool: MRP System

Format: Intranet

The MRP Library is the controlled collection of baselined items that comprise the baselined delivered product for production purposes. The MRP Library constitutes the master from which items are printed or duplicated. The MRP library provides storage of and controlled access to baselined items in a form suitable for printing, uploading, or network installation. The MRP components are documented in an identification list for each of the baselines. The Section 3.1.1 describes identification for each baseline collection.

The approved types 1, 2, 3, and 4 (as per EWI-006) documents are stored in the MRP library in Portable Document Format (PDF) such that their approval signatures appear on the approval page.

Distribution:

The DCG is responsible for distributing copies of the software to Production. Distribution to production occurs using the Controlled Document Release Form. Production is responsible for making and distributing copies of software products. The copies are made from the corporate MRP system.

The Production department is responsible to load each SCI (DCU Application, DCU Boot Loader Application, DCU Neuron Gateway Application and PTU Application) into the DCU as per the Door System Software Programming Procedure document #200910476. This document specifies the equipments (including the SCI P/N and revision number), the steps required to program every programmable device inside the DCU, and the method to verify that the SCIs have been correctly programmed.

The STL is responsible to generate the PTU Application Installation Package as well as the Installation Procedure. The PTU Application Installation Package is released onto CD-ROM Media along with the Installation Procedure. The installation of the PTU Application into the PTU Computer is not the scope of VSRS.

Verification:

The STL is responsible for review and approval of a software product. Once approved, the STL informs the DCG that the software product is ready for release. The SCM is responsible for ensuring that the correct software product and release documentation is distributed through the DCG.

Tracking:

Changes to approved baselined items in the MRP library are governed by the ECD process as documented in EWI-010 (part of the VSRS QAM). The ECD process is managed by the DCG per the VSRS QAM (outside the scope of this SCMP).

Security:

The MRP Library's write-access is limited to the DCG, while read-access is available to all the engineering personnel, and to selected Production personnel.

Retrieval:

A few stations are provided at the Production department to access the MRP library Intranet. Instructions to retrieve items from the MRP library are provided on the Intranet pages.

Printing:

Printing is achieved by opening the document with the associated tool, and printing to a printer.

Media Duplication:

For media duplication, a media programming procedure document exists for each media item (see Figure 3-2). The software constituents of a media (Electronic Document Files, Electronic Drawing Files, and Software Electronic Files) are directly pulled off the MRP library Intranet. Each procedure includes a media verification process.

3.2 Configuration Control

This section describes the process of configuration control of all identified SCIs developed or maintained.

The purpose of configuration control is to maintain the integrity of baselined SCIs and their associated documentation by ensuring that only authorized changes are incorporated. This requires the systematic evaluation, processing, and approval or disapproval of all proposed changes. Configuration control begins when an SCI is baselined and continues as further baselines are established.

SCM is responsible for maintaining software configuration control over software products in the Functional, Allocated, and Product baselines. In addition, SCM is responsible for administering the process by which a request for change to product under configuration control is submitted, reviewed, and approved or disapproved.

The following Figure describes the lifecycle of an SPR as well as the involvement of all those implicated in its correction.

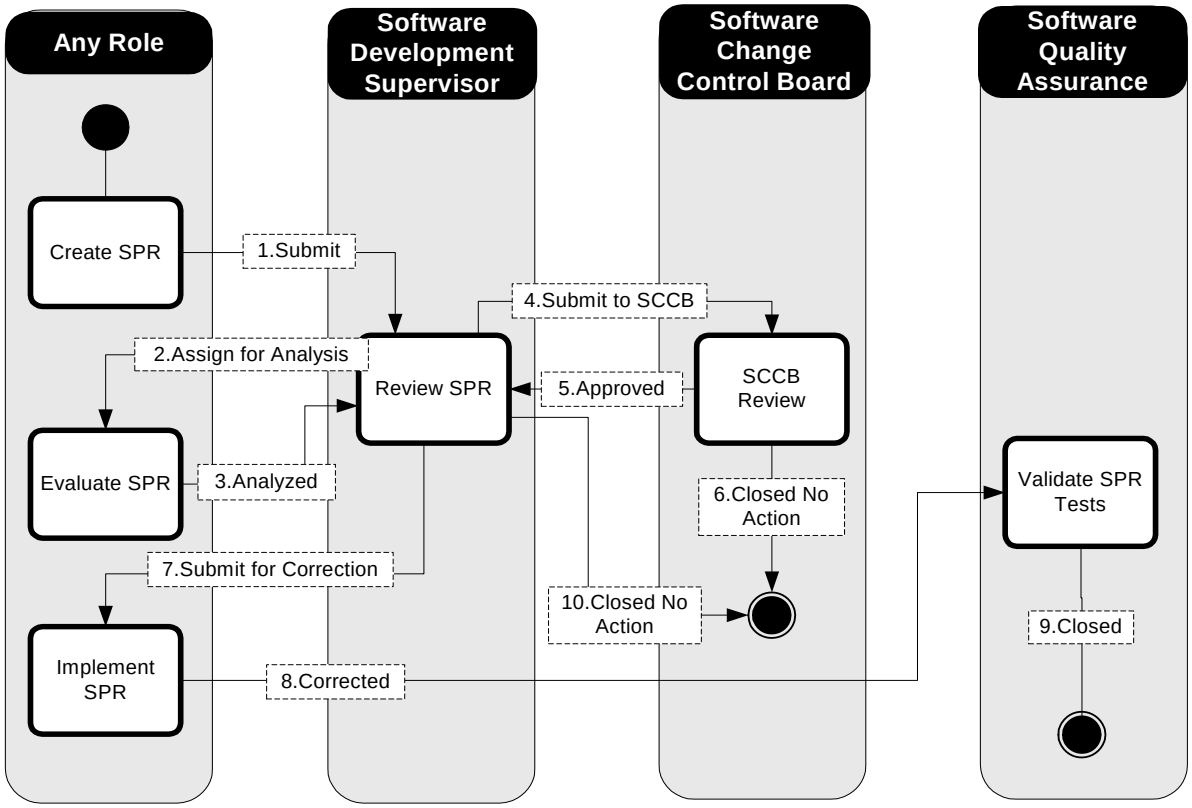

Figure 3-3: Configuration Control Process

3.2.1 Requesting changes

Changes to a baselined configuration are initiated through a change request process that involves the preparation of SPR form using the SPR Database tool. Note that no SPRs are submitted as long as the software is not baselined. The SPRs are used to report problems and propose changes or enhancements to software, system or documentation. An SPR must be documented, submitted, reviewed, and approved prior to implementation. Change requests against Developmental Baselines before a Product Baseline are resolved by the Project Manager and do not need the SCCB approval. Change requests against approved baselines require the approval of the SCCB.

VSRS uses the SPR Database for controlling changes. A description of the change classification definition and information are presented in the following sections.

When submitting an SPR, the following information is provided:

- SPR number (automatically assigned);
- The name(s) and version(s) of the Configuration Items (CIs) where the problem appears;
- Originator's name and organization;
- Date of request (automatically recorded);
- Indication of priority;
- Synopsis;
- Description of the requested change;
- Attachments (letters, test results, waivers).

Status is marked as *Submitted*.

3.2.2 Evaluating changes

A *Submitted* SPR is technically verified and a proposed solution is drafted. Changes are evaluated according to their effect on the SCIs and their impact on project resources.

When evaluating an SPR, the following information is provided:

- List of SCI affected;
- Textual description of the proposed solution;
- Estimated effort required to implement;
- Level of risk;
- Impact on software and system documentation (e.g. SFD, SRS, SDD, Test documents, and other relevant documents).

Once evaluated, the SPR status is marked as *Verified*.

3.2.3 Approving Changes

The following subsections specify the process for approving changes and the project members responsible for approving changes.

3.2.3.1 Change Approval

Once an Allocated, Functional, Developmental or Product baseline is established, all changes have to be approved. The changes related to the Developmental Configuration[1] are approved by the Program Manager. However, the changes related to the Allocated, Functional, and Product baselines require an approval from the SCCB. In order to obtain the SCCB approval, the SCM schedules a SCCB meeting with the personnel identified in the Table 2-2 to review the proposed disposition of the SPRs.

Once approved, the SPR status is marked as *Assigned* and an individual is assigned to implement the changes.

3.2.3.1.1 Software Configuration Control Board

For information regarding the SCCB, its level of authority and its responsibilities, refer to Section 2.2.1.1.

3.2.4 Implementing changes

3.2.4.1 Change Process

The assigned individual implements the approved changes and provides a status of the implementation and completion to the SCM. Implementation includes updating the software and configuration documents. Implementation is considered complete when the integration and testing of the change request *passes* test criteria.

At the completion of the implementation, the following information about the associated change request is recorded in the SPR Database:

- The name and version (or revision) of the affected item(s);
- List of documents modified, and revision;
- List of source files modified;
- The verification and validation data (including the test case identifier if required);
- Problem Source (e.g. Requirements, Design, Implementation, Testing).

Status is marked as *Resolved*.

After reviewing the test results and supporting documentation, the SQA closes the Software Problems Reports (SPRs) according to the SCCB's directives and modifies the status as *Closed*.

3.2.4.2 Release Process

The release of implemented changes occurs through the ECD process as per Work Instruction EWI-010, using the form 04-002. Note that the ECD process is only performed after the documents are baselined

Multiple approved changes may be released simultaneously.

[1] For changes subsequent to software delivery, the Program Manager holds also the responsibility to prepare, process, and obtain both car builder and transit authority approval of the FMI. The Software Team Leader gives support to the Program Manager in matters of technical nature.

3.3 Configuration Status Accounting

The Configuration Status Accounting is an element of configuration management, consisting of the recording and reporting of information needed to manage a configuration efficiently. Configuration status accounting will be performed as per SCM Guide (SWI-01).

3.3.1 SCI Status Accounting

For each software development project, status accounting activity shall be performed when the baseline of each SCI is established. The CM tools and SPR Database shall be used to collect, store, process, and report status accounting information. The type of information tracked for status accounting is as follows:

- Status of the SPR evolution (SPR process is described in Section 3.2).
- Versions and labels for source files

Prior to the baseline, no status accounting reports are generated to track the evolution of the SPR or the version of the software. Once a baseline is established, the information regarding the SPR evolution and the Software version is included in the SVD.

Access to the status data is governed by EWI-003 and managed by VSRS' Management Information Service group (outside the scope of this SCMP).

3.3.2 SCI Baseline Accounting

The information tracked for a baseline consists of a listing of the initial approved version (of each SCI), the current status and historic of proposed changes to the configuration, and the implementation status of approved changes. This information is gathered using the CM tools and SPR Database and is incorporated in an SVD.

3.4 Configuration Audits and Reviews

This section describes the approach used in performing configuration audits.

Configuration audits validate that the design and the final product conform to approved functional requirements defined in specifications and drawings and that the changes to the initially approved specifications and drawings have been incorporated.

The SCM assists in the conduct of two audits for developed baselines prior to their release: the FCA and PCA. These audits ensure that baseline changes are validated and the new baseline meets the requirements and specifications.

SCM provides assistance through the specific activities listed below, as required by the project:

- Assist in the audit;
- Review audit checklists;
- Prepare SCM reports, logs, or records required to support the audit;
- Establish and maintain baseline specification and product files;
- Follow up on audit reports to assess possible SCM impact;
- Provide storage for audit documentation, records, and products;
- Ensure audit report action items are resolved.

The audits to be performed by the SCM and the SQA are documented in the SQA Guide document #SWI-13 and the SCM Guide document #SWI-01. These documents provide also the schedule, the participants, the documentation required, the procedure for recording deficiencies, the approval criteria, and the specific action(s) to occur upon approval.

3.4.1 Formal SCM Audits

3.4.1.1 Functional Configuration Audit

SCM ensures that the released version of the software products is available for the audit so that the inspectors can verify that the software performs as required by its allocated configuration.

FCA is conducted after a major change or a significant number of minor changes have occurred or before the establishment of the Product Baseline. The FCA is scheduled in CMM-01 module SV-700. The SCM is responsible for assisting SQA in the preparation of the FCA plan. The FCA plan identifies specific tasks and procedures to accomplish those tasks. The FCA plan identifies documents, hardware, software, and test sets required for performing the audit. The SCM records differences between the SRS and SCI under audit for incorporation into the minutes of the FCA for post-audit action.

3.4.1.2 Physical Configuration Audit

This audit ensures that the as-built configuration is accurately reflected by the released documentation to establish the Product Baseline. SCM audits the released engineering documentation and quality control records to make sure the as-built or as-coded configuration is reflected by this documentation.

PCA is conducted concurrently with FCA or immediately following an FCA. The PCA is scheduled in CMM-01 module SV-700. The SCM is responsible for assisting SQA in the preparation of the PCA plan. The PCA plan identifies specific tasks and procedures to accomplish those tasks. The PCA plan also identifies the software and technical documentation to be examined.

3.4.2 In-Process SCM Audits

3.4.2.1 Configuration Management Audit

The Configuration Management audit ensures that the configuration management methods, defined in the SCMP, are respected throughout the course of the project. The Configuration Management audit shall be scheduled and performed according to CMM-01 module SV-700. This type of audit is performed by SQA and may be performed together with the FCA/PCA audits. Products generated or tracked by SCM are listed below:

- Identified SCIs;
- Change requests;
- Software version releases;
- Libraries;
- Documented SCM procedures;
- SCM review reports.

Audit findings are documented in an audit report and on Improvement Action Requests (IAR) forms S13-02 and provided to the SCM. The audit report is used by the SCM to correct deficiencies or identify changes in the SCM requirements. Correcting deficiencies would include updating SCM processes and procedures, records, configuration documents, software, or tools. Identifying changes in the SCM requirements would result in adding, modifying, or deleting a requirement in this SCMP.

3.4.2.2 Customer Reviews

The following audits and reviews related to Software Configuration Management are performed by the Car-Builder jointly with the Supplier:

- Functional Audit: Hyundai Rotem Project Engineering and Quality review the Supplier's SQAP document as part of the Software Documentation Assurance Procedure to ensure that the Supplier's Functional Audit procedures are both sufficient and correct.
- Physical Audit: Hyundai Rotem Project Engineering and Quality review the Supplier's SQAP document as part of the Software Documentation Assurance Procedure to ensure that the Supplier's Physical Audit procedures are both sufficient and correct.

- Software Configuration Management Plan Review: Hyundai Rotem Project Engineering reviews, approves and verifies the adequacy and the completeness of the configuration management methods contained in the Supplier's SCMP.

3.5 Interface Control

This section identifies the interface requirements and establishes the Interface Control Working Group. Interface management is performed to ensure compatibility and interoperability among various hardware and software components in a system, and to interfacing items outside the scope of the SCMP.

3.5.1 Dependencies

The software project interfaces with external and internal items to the system. The project SCIs are dependent on the Car network and the PTU hardware. Refer to Table 3-3 for further details on the external interfaces of the system. As for the internal dependencies of the SCIs, the DCU App., and the DCU Boot Loader App. interface with the PTU App. In case there are modifications to the DCU Application, the DCU Neuron Gateway Application, the DCU Boot Loader or the PTU Application, interfaces shall be verified for potential interfacing effects and vice versa. The software project may also be impacted by the mechanical and electrical components internal to the SCIs. For further details on the internal interfaces of the software project, refer to Table 3-4.

3.5.2 Interface Configuration Control

The configuration control of the internal and external interfaces are described in the Interface Control columns of Table 3-3 and. As for the groups involved in the configuration control of the interface items, refer to Interfacing Group(s) column of Table 3-3 and Table 3-4.

Table 3-3: System External Interfaces

System Interface	Interfacing Group(s)	Interface Control
Car Network Interface	Customer's System Engineering	The interface document is authored by Hyundai Rotem and released by Hyundai Rotem's Purchasing officer to VSRS' Program Manager through an official letter.
	VSRS Software Engineering	Document sent to Project Manager, reviewed, and then handled as per Section 3.2.
PTU Hardware Interface	Customer's System Engineering	The interface document is authored by Hyundai Rotem and released by Hyundai Rotem's Purchasing officer to VSRS' Program Manager through an official letter.
	VSRS Software Engineering	Document sent to Project Manager, reviewed, and then handled as per Section 3.2.

Table 3-4: System Internal Interfaces

Internal Interface	Interfacing Group(s)	Interface Control
DCU App. and DCU Boot Loader App. to PTU App. Interface	VSRS Software Engineering	As per Section 3.2.
SCIs. to Electrical and Mechanical Interface	VSRS Software, Electrical and Mechanical Teams	The software, electrical and mechanical issues are discussed among the managers and communicated to their teams. Once a baseline is established, these issues are addressed in the SCCB meetings.

3.6 Subcontractor and Vendor Control

This section is not applicable.

3.7 SQA Audit Activities

SQA Audit activities will be performed as defined in the SQAP document # 201111208.

4 SCM SCHEDULES

This section describes the software development activities specified in the CMM-01 for software-related groups and the SCM responsibilities in relation to the program events.

4.1 SCM Sequence and Dependencies

The SCM activities are defined within the Software Development Process module in CMM-01 and also in QSP-04 (Part of the VSRS' QAM). Refer to project's schedule in the SPMP for more information related to the SCM activities as well as the sequence and dependencies among all SCM activities.

4.2 SCM Milestones and Schedule

The SPMP document #201111207 defines the SCM milestones in relation to software-related group activities. The sequence of the activities to be performed as well as the completion criteria to reach these milestones is defined in the SDP module of CMM-01.

Refer to project's schedule in the SPMP for more information related to the SCM activities as well as the sequence and dependencies among all SCM activities.

5 SCM RESOURCES

This section identifies the software tools, techniques, equipment, personnel, and training required for the implementation of the SCM activities.

5.1 Tools

The IBM Rational ClearQuest Version: 7.1.1 database is used to control, track and monitor issues found after a software work product has been baselined or released. Software problem reporting and tracking is performed as per the *SWI-01: SCM Guide*.

The Microsoft Visual SourceSafe is used to maintain the Software Development Library. SCM is responsible for acquiring, installing, and maintaining the tool.

The project tracking database (ABAK) software is the corporate time tracking tool. It is used to track SCM effort.

IBM Rationale RequisitePro is used to trace and maintain the requirements and also to develop technical documentation that contain requirements.

Historical Folder shall be used to archive documentation. All electronic data is backed up according to QSP-S14-02 and managed by VSRS' Management Information Service group

5.1.1 Configuration Management of Tools

The configuration control of the tools listed in Section 5.1 is managed by VSRS' Management Information Service Group and is outside the scope of this SCMP.

5.2 Techniques

- CMM-01 (Software Engineering Process): The SEP is a framework for development of software projects at VSRS. The SEP follows rules, requirements, and activities required in the CMM for Software, level 2, from the Software Engineering Institute (SEI).

  The SEP establishes a common process framework that defines, implements, and supports the VSRS organization by specifying the methods, people and tools necessary to achieve projects and software products with the lowest cost, best delivery time and the lowest rates of defects.

  The SEP delineates the framework that shall be applied in any research, design, development, maintenance or acquisition of software products by VSRS. The SEP provides a tool for VSRS management to ensure that the people, products and processes involved, are supported by the organization and a set of methods and tools capable of meeting the standards of performance required by company objectives.

  VSRS personnel must apply a predictable engineering process to design, develop, subcontract or maintain software products.

- CMMI-02 (Software Engineering Policies): This policy manual establishes management expectations for developing and sustaining a CMMI Level 2 compliant software engineering process capability within VSRS.

- SWI-01 (Software Configuration Management Guide): This document describes the Software Configuration Management process used by VSRS to define and identify software baselines along with their components and documentation, to control changes to those baselines, to verify the conformance of those baselines to the original documentation and/or to the documented changes and to maintain records of the baselines status. This guide defines outlines to all users and ensures that different individuals can repeat a set of auditable activities with appropriate diligence.

- EWI-003 (Project Folder, Naming and Permission): This document describes a project folder structure and naming methodology for the Engineering folder.

- QSP-S14-02 (Data Storage, Preservation and Security): This document describes the methods to be used by VSRS to assure that all electronic documents are backed-up, maintained and preserved.

5.3 Methodologies

CMMI® for Development, Version 1.2, August 2006, Software Engineering Institute, Carnegie Mellon University.

BS EN 50128, Railway Applications Communications, Signalling and Processing Systems- Software for railway control and protection systems, Version 2001.

5.4 Personnel

The personnel required are identified in the SPMP Section 5.1.2. A detailed software schedule is also presented in appendix A of the SPMP.

5.5 Training

The training is identified in the SPMP Section 5.1.4.

5.6 Environment and Infrastructure

The equipment required for SCM is VSRS' servers and configuration tools listed in Section 5.1. For information on used techniques, refer to Section 5.2. For additional information regarding standards, procedures and infrastructure, refer to Section 2.3 and Section 3.1.

6 SCM MAINTENANCE

The maintenance of the SCMP consists of all the activities that are necessary to ensure that the SCM planning is performed continuously throughout the life cycle of the project.

6.1 SCMP Monitoring

The VSRS' SCM is responsible for monitoring and updating the SCMP to reflect the current SCM planning, processes, resources and standards.

6.2 SCMP Revision and Approval

The SCMP is revised upon a customer request or an internal assessment to modify the requirements related to standards, techniques and processes that have a direct impact on the configuration management.

Changes and reviews are handled as per Section 3.2. Also, the SCMP undergoes the inspection/approval process as per VSRS' SWI-05 prior to the release.

6.3 SCMP Change Distribution

Once the SCMP is updated and approved, it is placed under the appropriate library. The project team members are informed about the existence of the latest revision of the SCMP during the SPRM.